

Fig. 1: The IIoT stack from an automation perspective

1000 such controls are used during production, from the beginning of the manufacturing process to the point of dispatch.

The IIoT harnesses sensor data, machine-to-machine (M2M) communication and automation technologies. Smart machines are better than humans at accurately and consistently capturing and communicating data used to fix inefficiencies and solve problems in terms of up-time, scheduled maintenance, power efficiency and more efficient utilisation, sooner.

Maxim Integrated has broken down the IIoT in terms of a stack as shown in Fig. 1. At the very bottom of the IIoT stack, they have the devices (systems) on the factory or process floor. These can be field sensors, controllers, industrial PCs and so on. All of these are hardware systems and can include aspects of hardware security. These end devices must have useful data to communicate and are generally hooked up to communication hubs, gateways and switches so that data can be put in the Cloud (or an intranet) as Big Data.

But that is not all. The promise of the IIoT is that this data can be integrated within the ERP and CRM software of the firm to not only efficiently plan and cost out a manufacturing process, but also to use customer/market information to change assembly lines and process parameters.

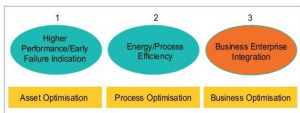


Fig. 2: Potential IIoT benefits

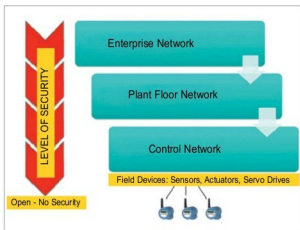


Fig. 3: Mapping security by plant network levels

The top of the stack impacts software development and integration, whereas the bottom impacts the system design perspective.

Primarily, the benefits of IIoT can be broken down into three groups (Fig. 2): asset, process and enterprise optimisation. It is easier to optimise a motor than it is to optimise a drilling operation, which, in turn, is easier to optimise than the manufacturing lines of a large enterprise. But optimising at every level is the dream of the IIoT.

The first level of analysis and interaction occurs at the edge—data is collected from a sensor (for example, a wind turbine sensor, a motor encoder or a vibration signature). This is processed locally to help understand how to tweak parameters that would give the highest efficiency or provide an early indicator of a potential failure.

The next level of analysis is done at the control room or plant level where sensor data from multiple end devices and even multiple assembly lines is aggregated to

make decisions that would increase the efficiency of the factory or a process; for example, a control room making idling or sleep decisions of various end devices to reduce the overall power profile of the process.

The first two aspects of using data to positively impact operations is what you are familiar with and use in some way, shape or form. However, what the IIoT envisions is not just an increase in data collection and analysis at the first two stages, but integrating process

data with enterprise data to make really interesting decisions that so far have not been made before.

Consider a company enjoying a market explosion. The assembly line can be programmed to manufacture higher volumes of the product, or completely bypass sub-assemblies adding features not valued by the market. Now, combine both the operating and financial data to provide more insight to the chief financial officer. The agility of the company and its ability to pivot, change and continue to grow can be exponential. Indeed, it is an attractive proposition, and many are eager to move forward, quickly. So quickly that security has not been keeping up with the new IIoT systems.

The IIoT exposes system vulnerabilities

There are a few ways that IIoT systems are vulnerable to attacks. Among the two most prominent are Cloud storage and network architecture.